

99. B. A. Zaitsev and A. I. Grivkova, Radioactive Cesium-137 [in Russian], Gosatomizdat, Moscow (1961).
100. B. A. Zaitsev and T. S. Shevelkina, Radiokhimiya, 12, 860 (1970).
101. V. I. Spitsyn, P. N. Kodochigov, M. M. Golutvina, A. F. Kuzina, and Z. A. Sokolova, Methods for Work with Radioactive Indicators [in Russian], Izd. Akad. Nauk SSSR, Moscow (1955).
102. V. I. Spitsyn and E. A. Torchenskova, Dokl. Akad. Nauk SSSR, 95, 289 (1954).
103. V. I. Spitsyn and Yu. I. Bykovskaya, Dokl. Akad. Nauk SSSR, 104, 256 (1955).
104. V. I. Spitsyn, V. D. Balukova, A. F. Naumova, V. V. Gromov, and S. I. Zakharov, Proceedings of the Second UN International Conference on the Peaceful Uses of Atomic Energy, Vol. 18, UN, Geneva (1958), p. 439.
105. V. I. Spitsyn and V. V. Gromov, Pochvovedenie, No. 12, 45 (1959).
106. V. V. Gromov and V. I. Spitsyn, Man-Made Radionuclides in the Marine Environment [in Russian], Atomizdat, Moscow (1975).
107. V. I. Spitsyn, G. P. Bondarenko, V. V. Gromov, and A. V. Vershinin, Dokl. Akad. Nauk SSSR, 252, 1183 (1980).

OXIDIZING AGENTS PRODUCED BY RADIOLYSIS OF ALKALI-METAL HALIDE MELTS

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The principal short-lived products from the radiolysis of alkali-metal halide melts [1, 2] are solvated electrons (e_s^-), which are particles with the F-center type structure [2], and molecular anion-radicals of the halogen (Γ_2^-); these are formed in the reactions [1]



Solvated electrons are strong reducing agents, and they disappear almost completely in reactions with the metal cations [2, 3]. Here we are presenting results from a study of the properties of the oxidizer products Γ_2^- , by means of a pulse radiolysis method [1, 2].

EXPERIMENTAL

The pulse radiolysis unit used in this work consisted of a linear electron accelerator U-12 (electron energy 5 MeV, current in pulse up to 0.2 A, pulse length 2.3 μ sec) and a fast-acting spectrophotometric system providing operation in the spectral range of 300-1600 nm, with a time resolution of 0.2 μ sec and a spectral resolution from 5 nm in the 300-400 nm region to 15 nm in the 800-1200 nm region. The lower limit of the spectral range is determined by the inherent absorption of the melt (in the present case, chlorides); for the bromides and iodides, this limit is shifted in the long-wave direction (to 350 and 400 nm, respectively). The procedures used in irradiating and preparing the samples have been described in [1, 2]. In studying the temperature dependence of the rate of reactions of Γ_2^- over a broad range of temperatures, we used melts of mixtures of NaCl and CsCl (34/66 mole %, mp 493°C) and LiBr and KBr (60/40 mole %, mp 348°C).

The kinetics of disappearance of Γ_2^- with assigned values of the process parameters were modeled in an MN-7M analog computer.

DISCUSSION OF RESULTS

The Cl_2^- ion-radicals in the chloride melts absorb in the violet and UV regions of the spectrum. For the Br_2^- and I_2^- in melts of bromides and iodides, this absorption is accompanied by considerably less intense absorption in the long-wave part of the spectrum (Fig. 1).

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TABLE 1. Parameters of Optical Spectra of Halogen Anion-Radicals in Alkali-Metal Halide Melts (for comparison, parameters are listed for the spectra of Γ_2^- in aqueous solutions [4, 5])

Γ_2^-	System	T., °C	$\lambda_{\max}$, nm	$W_{1/2}$, eV	ϵ_1 , liters/ (mole · cm)	ϵ_{II}/ϵ_I *
Cl_2^-	$10^{-3} M \text{ Cl in H}_2\text{O}$	25	340	0.93	8800	—
	LiCl	700	360	1.1 ± 0.3	7400	—
	NaCl	850	365	1.2 ± 0.3	6800	—
	KCl	800	360	1.25 ± 0.2	6500	—
	RbCl	750	370	1.3 ± 0.2	6300	—
	CsCl	700	370	1.4 ± 0.3	5800	—
Br_2^-	$10^{-3} M \text{ Br}^- \text{ in H}_2\text{O}$	25	360(I) 700(II)	0.93	8300	0.041
	NaBr	760	420(I) 725(II)	1.15 ± 0.3	6700	0.2 ± 0.1
	KBr	800	420(I) 750(II)	1.3 ± 0.3	5900	0.2 ± 0.1
	CsBr	680	410(I) 750(II)	1.2 ± 0.3	6400	0.2 ± 0.1
	$10^{-3} M \text{ I}^- \text{ in H}_2\text{O}$	25	375(I) 725(II)	— †	11560	0.27
		700	450(I) 750(II)	— †	10700	0.5 ± 0.1

* Ratio of molar extinction coefficients at maxima of long-wave (II) and principal (I) absorption bands of Br_2^- and I_2^- .

† Difficult to estimate half-width of main absorption band of I_2^- in melt, because of significant overlapping with additional long-wave band.

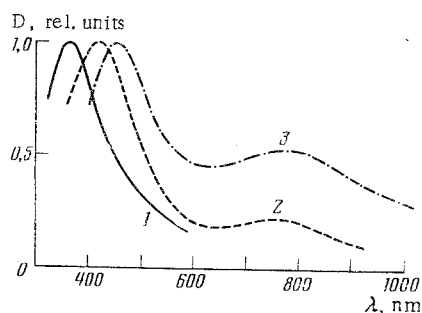


Fig. 1. Optical absorption spectra of Cl_2^- (1), Br_2^- (2), and I_2^- (3) in melts of potassium halide: 1) KCl (800°C); 2) KBr (800°C); 3) KI (700°C).

The optical absorption spectra of the Γ_2^- ion-radicals in the alkali-metal halide melts (Table 1) are similar to the spectra of the corresponding species in aqueous solutions [4, 5]. In melts, however, they are slightly shifted toward the long-wave region; this is related not only to the different nature of the solvent (ionic liquid), but also to the higher temperatures. The influence of these factors should be quite significant, since in the process of electronic excitation, the Γ_2^- ion-radicals behave as charge-transfer complexes. This, apparently, is also responsible for the higher relative intensity of the second long-wave absorption band of Br_2^- and I_2^- in comparison with the aqueous solutions (Table 1). This phenomenon has been noted for Br_2^- in melts of tetrabutylammonium bromide [6], where the ratio of intensities of the long-wave and principal bands was found to be 1/5.

The disappearance of Γ_2^- is not accompanied by any appreciable change in the observed spectrum; this indicates the absence of any superimposition of spectra of the products from the reactions taking place, and hence we can evaluate the reaction kinetics quite reliably on the basis of the change in optical density. The disappearance of e_s^- is due mainly to reduction of metal cations



The subsequent oxidation of the reduced form of the cation should be the most effective channel for the disappearance of the Γ_2^- ion-radicals. The view that there is practically no direct reaction between e_s^- and Γ_2^- is supported mainly by the substantial difference in lifetimes (τ) of these species when they are present at identical initial concentrations; see the reaction equations (1)–(3). For example, if we define the lifetime τ as that time in which the initial concentration drops by 90%, the value for e_s^- is no greater than 2 μsec , whereas for Cl_2^- , Br_2^- , and I_2^- , the respective average values of τ are 5, 50, and 500 μsec .

TABLE 2. Rate Constants of Reactions $\Gamma_2^- + M_2^{(2n-1)+}$, Calculated from Drop in Optical Absorption of Halogen Anion-Radicals

Γ_2^-	M	System	T., °C	k, liters/(mole · sec)
Cl_2^-	Na	NaCl	850	$(2.0 \pm 0.2) \cdot 10^{11}$
	K	KCl	800	$(1.0 \pm 0.3) \cdot 10^{11}$
	Rb	RbCl	750	$(6.7 \pm 0.2) \cdot 10^{10}$
	Na (Cs)	NaCl—CsCl	550	$(3.6 \pm 0.5) \cdot 10^{11}$
		(34/66 mole %)	600	$(4.0 \pm 0.7) \cdot 10^{11}$
			700	$(3.7 \pm 0.8) \cdot 10^{11}$
			750	$(4.0 \pm 0.9) \cdot 10^{11}$
			800	$(4.4 \pm 0.9) \cdot 10^{11}$
			850	$(4.8 \pm 0.5) \cdot 10^{11}$
	Br_2^-	Cd	NaCl	850
K		KBr	800	$(4.0 \pm 0.4) \cdot 10^{10}$
Cs		CsBr	600	$(4.2 \pm 0.5) \cdot 10^{10}$
Li (K)		LiBr—KBr	400	$(2.4 \pm 0.7) \cdot 10^{10}$
		(60/40 mole %)	500	$(2.3 \pm 0.9) \cdot 10^{10}$
			600	$(2.9 \pm 0.4) \cdot 10^{10}$
			700	$(2.2 \pm 0.4) \cdot 10^{10}$
			800	$(2.2 \pm 0.2) \cdot 10^{10}$
			850	$(2.3 \pm 0.4) \cdot 10^{10}$
Cd		KBr	800	$(0.7 \pm 0.3) \cdot 10^{11}$
Zn		KBr	800	$(0.7 \pm 0.3) \cdot 10^{11}$
Cd		LiBr—KBr	400	$(3.6 \pm 0.4) \cdot 10^{10}$
		(60/40 mole %)	500	$(4.6 \pm 0.4) \cdot 10^{10}$
			600	$(6.4 \pm 0.5) \cdot 10^{10}$
			700	$(0.9 \pm 0.2) \cdot 10^{11}$
			800	$(1.0 \pm 0.2) \cdot 10^{11}$
			850	$(0.9 \pm 0.5) \cdot 10^{11}$
I_2^-	K	KI	700	$(3.8 \pm 0.4) \cdot 10^9$

The decrease in the observed rate constant for the disappearance of e_s^- with increasing temperature was explained in [3] by the reversibility of reaction (4). However, the reversible reaction (4) is apparently followed by the practically irreversible reaction



the product of which, $M_2^{(2n-1)+}$ is oxidized by the Γ_2^- ion-radical



If $k_6 \cdot [\Gamma_2^-] \cdot [M_2^{(2n-1)+}] \ll k_5 \cdot [M^{n+}] \cdot [M^{(n-1)+}]$, the processes (4)-(6) prove to be separated in time, and the disappearance of Γ_2^- will be described by a hyperbolic relationship, in accordance with the equation

$$d[\Gamma_2^-]/dt = k_6 \cdot [\Gamma_2^-] \cdot [M_2^{(2n-1)+}] = k_6 [\Gamma_2^-]^2 \quad (7)$$

This relationship is actually observed in the pure matrix, where the alkali metal enters in as M, and it is also observed in the presence of cations of d-elements (Fig. 2). On the basis of the linear anamorphoses of second-order reactions, rate constants were obtained (Table 2); in calculating these rate constants, we used values of the Γ_2^- extinction coefficients in aqueous solutions [4, 5], corrected for broadening of the absorption bands and redistribution of the oscillator force owing to the increase in relative intensity of the long-wave band. Here we must turn attention to the following facts:

- 1) The rate constants of the reactions of Cl_2^- are approximately an order of magnitude greater than the corresponding constants for Br_2^- , which in turn are an order of magnitude greater than the constants for I_2^- .
- 2) The rate constants for the reactions of Γ_2^- with an alkali metal are practically independent of temperature.
- 3) The rate constant for the reaction of Br_2^- with Cd^{2+} increases with increasing temperature, this increase corresponding to an activation energy of approximately 5 kcal/mole, close to the activation energy of self-diffusion [7].

The presence of Group II s-elements in the melt leads to a complication of the kinetics observed for the disappearance of Γ_2^- . In this case, we find two very clearly expressed sections on the kinetic curve, i.e., a fast and a slow drop in concentration; this is apparently related to disproportionation



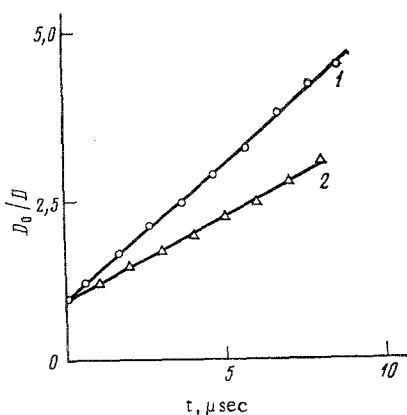


Fig. 2

Fig. 2. Kinetic curves for disappearance of Br_2^- in linear anamorphoses of second-order reaction. System: melt of mixture of LiBr and KBr (60/40 mole %) at 400°C . 1) With added Cd^{2+} (reaction of Br_2^- with reduced form of cadmium ion); 2) without additive (reaction of Br_2^- with alkali metal).

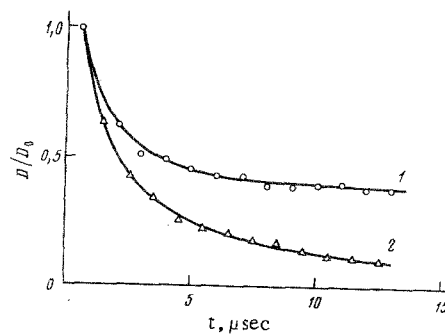


Fig. 3

Fig. 3. Curves for disappearance of Γ_2^- , calculated on the basis of Eq. (10) and (11) with assigned values $k_6 = 7.0 \cdot 10^{10}$, $k_9 = 1.5 \cdot 10^9$, and $k_8 = 7.0 \cdot 10^{10}$ liters/(mole·sec) (1) and $5 \cdot 10^9$ liters/(mole·sec) (2). The points on the curves represent experimental values obtained in KBr melts at 800°C in the presence of Ca^{2+} (1) or Cd^{2+} (2).

Thus, if $k_8 \gg k_6$, we find that during the process there is a sharp decrease in the ratio of concentrations of $\text{M}_2^{(2n-1)+}$ and Γ_2^- , as a result of which the rate of reaction (6) drops below the rate of the relatively slow reaction (9).

On the slow section of the kinetic curve for Br_2^- in a KBr melt with added Ca^{2+} , Ba^{2+} , or Sr^{2+} at 800°C , where linearity is observed on coordinates of $1/D$ vs t , we find that $k_9 = (1.5 \pm 0.5) \cdot 10^9$ liters/(mole·sec). Let us note that one possible explanation for the convergence of the rates of reactions (6) and (8) is the existence of Br_2^- in the form of a pair with an ion of the alkaline-earth element.

In order to estimate the value of k_8 , the scheme of the reactions taking place was modeled in an analog computer on the basis of the system of differential equations

$$-d[\Gamma_2^-]/dt = k_6[\Gamma_2^-] \cdot [\text{M}_2^{(2n-1)+}] + k_9[\Gamma_2^-]^2 \quad (10)$$

$$-d[\text{M}_2^{(2n-1)+}]/dt = k_6[\Gamma_2^-] \cdot [\text{M}_2^{(2n-1)+}] + k_8[\text{M}_2^{(2n-1)+}]^2 \quad (11)$$

In solving this system, the initial concentrations of Γ_2^- and $\text{M}_2^{(2n-1)+}$ were assigned as equal to the experimentally determined initial concentration of Γ_2^- (at the end of the pulse of ionizing radiation); the value of k_9 was found experimentally in the KBr melt at 800°C , $1.5 \cdot 10^9$ liters/(mole·sec); the value of k_6 was taken as equal to the corresponding value for KBr melts at 800°C with added Cd^{2+} , $7 \cdot 10^{10}$ liters/(mole·sec), since the initial slopes of the kinetic curves for Γ_2^- in melts with added Cd^{2+} and Ca^{2+} are approximately identical; the value of k_8 was varied over the interval from 10^8 to 10^{11} liters/(mole·sec). Good agreement with the experimental curve for the disappearance of Br_2^- in the presence of Ca^{2+} is obtained with a value $k_8 = (7 \pm 2) \cdot 10^{10}$ liters/(mole·sec). If $k_8 \leq 5 \cdot 10^9$ liters/(mole·sec), the dependence of $[\Gamma_2^-]$ on t that is obtained is very nearly hyperbolic, with a constant that corresponds to the assigned value of k_6 ; i.e., it coincides with the experimental curve for melts with added Cd^{2+} (Fig. 3). From these data it follows that the reduced form of the alkaline-earth metal ions has a greater tendency to disproportionate than does the reduced form of d-element cations. This difference may be explained on the basis that the d-element cations in melts exist in the form of complexes with the halide ions.

Thus, an analysis of the set of reactions (1)–(9) shows that in the radiolysis of alkali-metal halide melts, we typically find almost complete reconstruction of the original substance, even at the level of radical reactions. Radiolytic decomposition of the substance due to removal of halogen, for example in the gas phase, will increase when alkaline-earth metal ions are present, these ions promoting the occurrence of reaction (9). The molecular halogen formed as a result of reaction (9) can also participate in oxidation reactions; however, the substantially lower rate of such reactions in comparison with the reactions of radical species should bring about accumulation of the halogen in solution in significant concentrations.

CONCLUSIONS

1. The optical spectra of the oxidizer components of radiolysis of alkali-metal halide melts, the molecular anion-radicals Γ_2^- , are similar to the spectra of the corresponding species in aqueous solutions.
2. Rate constants have been measured for the disappearance of Γ_2^- , which goes forward mainly in the reaction of Γ_2^- with the reduced form of the cation.
3. In the presence of alkaline-earth metal cations, owing to the fast disproportionation of their reduced form, there is a greater contribution from the relatively slow reaction of disproportionation of Γ_2^- , leading to an observed retardation of the rate of Γ_2^- disappearance and an increase in the yield of molecular halogen.

LITERATURE CITED

1. I. E. Makarov, T. N. Zhukova, and A. K. Pikaev, *Radiat. Effects*, **22**, 71 (1974).
2. I. E. Makarov, T. N. Zhukova, and A. K. Pikaev, *Dokl. Akad. Nauk SSSR*, **225**, 1103 (1975).
3. T. N. Zhukova, I. E. Makarov, and A. K. Pikaev, *Dokl. Akad. Nauk SSSR*, **261**, 409 (1981).
4. L. I. Grossweiner and M. S. Matheson, *J. Phys. Chem.*, **61**, 1089 (1957).
5. J. H. Baxendale, P. Sharpe, and M. D. Ward, *Int. J. Radiat. Phys. Chem.*, **7**, 587 (1975).
6. I. E. Makarov, T. N. Zhukova, and A. K. Pikaev, *Khim. Vys. Energ.*, **15**, 402 (1981).
7. J. Bockris and G. Hooper, *Discuss. Faraday Soc.*, **32**, 218 (1961).

USE OF RADIATION IN STUDYING MECHANICALLY INDUCED DEFECTS IN POLYCRYSTALLINE SPECIMENS OF MAGNESIUM OXIDE

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Mechanical working of solids in various types of milling equipment leads not only to size reduction, but also to the formation of structural defects. Here we are reporting on a study of the structural defects formed in polycrystalline specimens of MgO by mechanical working in a vibratory mill (pressure and shear type of action). According to [1, 2], with such an action, defects are formed primarily in the surface layer of the grain; distortion of symmetry of the immediate surroundings of the cation is observed, as well as an increase in the specific surface area of the specimens. The nature of the mechanically induced defects in polycrystalline MgO specimens has not been studied adequately.

Since the vibratory dispersion of MgO has been found to increase the lattice constants, we can assume that this is related to a displacement of the ions in the equilibrium position and the formation of Schottky or Frenkel defects. An analogous phenomenon was observed in the irradiation of MgO with neutrons or protons [3], and also in plastic deformation of single crystals [4]. In order to identify the cationic and anionic vacancies in the mechanically worked MgO specimens, we used γ -irradiation with subsequent ESR analysis of the paramagnetic centers (PC) that were formed, in the γ -irradiation of MgO, radiation centers are formed only as a result of filling of "biographical" vacancies by electrons and holes [3, 5].

EXPERIMENTAL

Two types of commercial MgO samples were used in this work, differing in the content of impurities (Table 1) and in the specific surface area. Sample 1 was produced by the Kali Combine in the German Democratic Republic (East Germany); sample 2 was produced by Merck.

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